

CSA07 – Computer Networks

Assessment Tool 2 – Case Study: NimbusEdge Technologies Network Outage (March 18, 2024)

Answers to Q1–Q4, based on the Incident Report, Network Topology Diagram, NOC Change Log, and Client SLA Breach Notice

Q1. RIP — Chennai HQ Campus

PROBLEM STATEMENT

During a Delivery department onboarding project, a chain of temporary VLAN gateways pushed a newly provisioned subnet beyond RIP v2's 15-hop limit. The route was silently withheld from the routing table, and the fault was invisible to standard monitoring until client tickets forced discovery on the morning of the incident.

A) UNDERSTANDING OF THE CASE

Per the NOC Change Log, CHG-0221 (2024-02-05) added a temporary VLAN gateway chain at Chennai HQ to support a client onboarding project, requested by the Delivery IT Lead and approved by the Network Supervisor at Medium risk. CHG-0224 (2024-02-09) then provisioned the new Delivery subnet behind that gateway chain. Neither change was flagged as high-risk, and neither included an explicit hop-count check against Chennai HQ's RIP v2 domain.

The Incident Report's timeline shows the consequence: at 09:42 IST on March 18, the Delivery team raised the first ticket, unable to reach the newly provisioned subnet. At 10:15 IST, NOC engineers confirmed the subnet simply did not appear in the HQ core switch's routing table — not that it was reachable-but-slow, but absent entirely. The key issues are therefore: (i) the causal change — a chain of temporary VLAN gateways stacked in front of the new subnet; (ii) the technical trigger — the cumulative hop count from that subnet to the core switch exceeding RIP v2's usable range; and (iii) the detection gap — because RIP treats an over-limit route as 'unreachable' rather than as a fault, no alarm fired, and the problem surfaced only through end-user (client) impact rather than infrastructure monitoring.

B) APPLICATION OF CONCEPTS

RIP v2 is a distance-vector protocol that uses hop count as its only metric, with a maximum usable value of 15 hops. Each router that advertises a route increments the hop count by 1 before passing it to its neighbour. A destination reported at hop count 16 is defined by the protocol as unreachable — this is RIP's built-in 'infinity' value, originally chosen to bound the count-to-infinity problem in distance-vector loop scenarios.

Applying this step by step to the Delivery subnet: the temporary VLAN gateway chain inserted a series of additional Layer-3 hops between the subnet and the HQ core switch. As the route for that subnet propagated back toward the core, each intermediate gateway added 1 to its hop count. By the time the advertisement reached the 16th hop, RIP no longer treated it as a valid, reachable route — it assigned it metric 16 (unreachable) and excluded it from the routing table. Downstream of that point, the core switch had no route to the subnet at all, which is exactly what the NOC engineers observed at 10:15 IST: the subnet was missing from the routing table, not merely slow to converge.

C) ANALYSIS

No automatic alert was generated because, from RIP's point of view, metric 16 is not an error state — it is the protocol working exactly as specified. RIP has no concept of 'this destination has become unreachable due to a topology change I should flag'; it simply stops advertising/installing routes beyond the hop ceiling, silently. Standard infrastructure monitoring (interface up/down, device reachability, link utilization) saw nothing wrong: every physical link and every gateway device in the temporary chain was up and functioning normally. The fault existed purely at the routing-table level, which most monitoring stacks do not continuously validate at the level of 'is this specific subnet present in the table'.

This exposes a structural limitation of distance-vector protocols like RIP, not a one-off human error: RIP has no topology awareness (unlike link-state protocols, it does not know the overall shape of the network, only what its neighbours tell it), and its hop-count ceiling is an artificial scalability limit rather than a reflection of real link quality or capacity. Any sufficiently long chain of routers — however each individual hop was added for good, incremental reasons — can

silently exceed that ceiling with no single change looking risky in isolation. This connects directly to the monitoring gap the RCA report notes: NimbusEdge's monitoring watched device and link health, but not routing-table completeness or hop-count budget per subnet, so a purely protocol-level failure mode had no corresponding alert.

D) SOLUTION DESIGN

Two complementary solutions are appropriate:

- **Redesign the addressing/subnetting scheme:** collapse the temporary VLAN gateway chain and re-home the Delivery subnet no more than 2–3 hops from the HQ core switch, restoring comfortable headroom under RIP's 15-hop ceiling. Treat any future VLAN gateway insertion as a change that must be checked against the current maximum hop count on the Chennai HQ campus before approval, since the change log shows no such check was performed for CHG-0221/CHG-0224.
- **Protocol Migration:** Where flattening is impractical for future onboarding projects, migrate the Chennai HQ campus segment to a protocol without RIP's hop-count ceiling (e.g., EIGRP or OSPF), consistent with the migration already under way on the core backbone. This removes the structural limit entirely rather than only managing around it.

RIP remains a reasonable original choice for the Chennai HQ campus specifically: it is a small, stable topology (Finance, HR, and Delivery subnets on one campus), and RIP's simplicity, low configuration and operational overhead, and adequacy for small diameter networks make it a sensible fit when the network stays within a few hops. The March 2024 incident does not indicate RIP was the wrong protocol for HQ in general — it indicates that an ad hoc chain of temporary gateways, added incrementally for a specific project without a hop-count review, pushed a normally small network past the boundary within which RIP's design assumptions hold.

E) CLARITY

Problem: A stacked chain of temporary VLAN gateways pushed the new Delivery subnet to 16 hops from the core switch.

Reasoning: RIP v2 assigns metric 16 to any route beyond its 15-hop limit and treats it as unreachable — a designed protocol behaviour, not a fault condition, so no monitoring alert fired.

Recommendation: Flatten the gateway chain to restore hop-count headroom, add a hop-count check to the change process, and consider a hop-limit-free protocol for future growth — while recognising RIP was, and remains, appropriate for HQ's small, stable core.

Q2. IGRP — Bengaluru–Pune–Hyderabad Regional Link

PROBLEM STATEMENT

An ISP-side fault degraded the direct Bengaluru–Pune link, yet IGRP continued to prefer that degraded path over a healthier alternate route for an extended period, producing severe latency for Bengaluru-based client applications.

A) UNDERSTANDING OF THE CASE

The Network Topology Diagram shows a triangular regional interconnect: Bengaluru–Pune direct (2 Mbps), Hyderabad–Pune (8 Mbps), and Bengaluru–Hyderabad, all running IGRP inherited from a 2023 acquisition. The NOC Change Log records the origin of the fault: CHG-0229 (2024-02-14) logged the ISP being notified of intermittent fiber degradation on the Bengaluru–Pune segment, and CHG-0252 (2024-03-15) confirms the ISP-side fault was ongoing and worsening just days before the outage.

On the incident day, the RCA/Incident Report timeline shows the business impact directly: at 11:05 IST, the Bengaluru support desk reported severe latency (600 ms+) on applications routed via the Bengaluru–Pune link, while Hyderabad — reachable via the healthier Pune–Hyderabad (8 Mbps) leg — reported normal performance at that time. A structurally healthier alternate path (Pune–Hyderabad–Bengaluru) existed throughout, but IGRP did not shift client traffic onto it promptly, so Bengaluru users bore degraded performance for a sustained period until engineers manually adjusted route preference at 18:10 IST.

B) APPLICATION OF CONCEPTS

IGRP computes a composite metric from bandwidth and delay (with optional reliability/load terms, unused here). In its default form the metric is proportional to a bandwidth component — **10,000,000** divided by the minimum bandwidth in kbps along the path — plus a cumulative delay component; lower is preferred.

Path	Min. Bandwidth	Bandwidth Component (10,000,000 ÷ kbps)	Effect of Rising Delay
Direct: Bengaluru–Pune (degraded)	2,000 kbps	≈ 5,000	Delay term rises further as the ISP fault worsens, pushing total metric even higher
Alternate: Pune–Hyderabad– Bengaluru	8,000 kbps (<i>limited by smaller of two legs</i>)	≈ 1,250	Delay term stays near baseline since both legs are healthy

Even before the delay term is added, the bandwidth component alone makes the alternate path roughly four times more favourable than the degraded direct link. Once the ISP fault increases delay and reduces effective throughput further on the direct Bengaluru–Pune segment, the gap widens further in the alternate path's favour. Correctly functioning IGRP, recalculating on current link conditions, should therefore have converged on Pune–Hyderabad–Bengaluru as the preferred path well before 18:10 IST — the metric math clearly supported the switch.

C) ANALYSIS

The incident is a convergence-speed issue, not a metric-calculation issue. The composite-metric formula itself was capable of identifying the alternate path as superior once fed accurate bandwidth and delay figures — the numbers in part (b) show that clearly. What lagged was IGRP's reaction time: as a classic distance-vector protocol, IGRP relies on periodic (not event-driven) updates (roughly every 90 seconds by default) and uses holddown timers after a route change is detected, specifically to suppress rapid route flapping and prevent routing loops. That same loop-prevention mechanism that makes IGRP safe also makes it slow to react to a real, sustained degradation — it does not distinguish between transient noise and a genuine, lasting fault, so it holds down the old preferred path for a period that, in this incident, stretched into hours of Bengaluru-facing latency.

D) SOLUTION DESIGN

Two options are viable, and both are justified against NimbusEdge's regional network scale (three-site, moderate-bandwidth interconnect inherited from an acquisition, already scheduled for consolidation):

- **Tune IGRP timers:** (shorter update interval, reduced holddown period) to shrink the reconvergence window — a low-effort interim fix, but it only reduces the delay, it does not remove the structural cause.
- **Migrate the regional link to a faster-converging protocol:** EIGRP (event-driven updates via DUAL, with a pre-computed feasible successor for near-instant failover) or fold it into the OSPF migration already under way on the core backbone. This is the stronger long-term choice, since it directly targets the convergence-speed weakness rather than only masking it.

IGRP's original advantage over RIP was precisely the composite bandwidth-and-delay metric demonstrated in part (b): unlike RIP's flat hop count, IGRP could distinguish a 2 Mbps link from an 8 Mbps link and route around genuine capacity differences on a regional network with variable link bandwidths. That advantage did not fail in this incident — the metric calculation was sound. What failed was the protocol's reaction speed to a live degradation event, which periodic-update, holddown-based distance-vector protocols are inherently slower to handle than event-driven alternatives.

E) CLARITY

Metric calculation: Direct Bengaluru–Pune path **≈ 5,000** (bandwidth component alone, on a 2 Mbps link) versus the Pune–Hyderabad–Bengaluru alternate **≈ 1,250** (on the 8 Mbps-limited path) — the alternate was clearly preferable on bandwidth alone, before even adding the degraded link's rising delay.

Reasoning: The metric math favoured the alternate path throughout; the failure was IGRP's slow, hold-down-gated reconvergence, not an incorrect metric.

Recommendation: Tune IGRP timers as a short-term mitigation, and migrate the regional interconnect to a faster-converging protocol (EIGRP or OSPF) as the durable fix.

Q3. OSPF — Core Backbone Migration

PROBLEM STATEMENT

A transcription error during a phased OSPF rollout gave the Hyderabad router the wrong Area ID, preventing it from ever forming a neighbour adjacency with the Chennai Area 0 backbone router — creating a routing black hole that stayed invisible to standard monitoring for ten days.

A) UNDERSTANDING OF THE CASE

The Network Topology Diagram shows the backbone mid-migration to multi-area OSPF: Area 0 at the Chennai HQ backbone router, with Bengaluru, Pune, and Hyderabad each assigned to a separate regional area. The NOC Change Log traces the rollout: CHG-0233 enabled OSPF/Area 0 on the Chennai backbone router; CHG-0238 and CHG-0241 configured Bengaluru (Area 1) and Pune (Area 2) with adjacency explicitly verified for both. CHG-0245 (2024-03-08), by contrast, configured the Hyderabad router for its area but is flagged 'Implemented — Error Introduced': an incorrect Area ID was entered during that configuration, and — unlike the Bengaluru and Pune entries — its description does not record adjacency verification.

The Incident Report traces the consequence: at 12:30 IST on March 18, engineers attempting to reroute Hyderabad traffic during regional troubleshooting discovered the Hyderabad regional router had no adjacency with the Chennai backbone router, and traceroutes showed traffic being silently dropped. Because the physical link itself remained up throughout, this black hole went undetected from the March 8 misconfiguration until it was found mid-incident on March 18 — ten days of latent risk that surfaced only once Hyderabad traffic actually needed the backbone path during the outage.

B) APPLICATION OF CONCEPTS

OSPF routers can only form a full neighbour adjacency across a link when several parameters match exactly on both ends: Area ID (for that interface), Hello and Dead interval timers, authentication type/credentials, and subnet mask (for the shared network). A mismatch on any one of these blocks the neighbour relationship from progressing to the Full state — the routers may still exchange Hello packets and briefly see each other, but they will not exchange Link-State Advertisements or install routes learned from each other.

Here, the violated requirement is Area ID: CHG-0245 recorded an incorrect Area ID entered for the Hyderabad router, so its OSPF process did not match what the Chennai Area 0 backbone router expected for that link. That single mismatch is sufficient by itself to prevent adjacency formation, regardless of every other parameter being correct — Hello/Dead timers, authentication, and subnet mask matching would not compensate for an Area ID mismatch. The result is precisely what the Incident Report describes: no adjacency, no LSA exchange with the backbone, and therefore no route for Hyderabad-bound traffic across that link, even though the interface and physical link stayed up.

C) ANALYSIS

This fault was harder to detect than a physical link failure because it lived entirely at Layer 3 routing-protocol state, not at Layer 1/2. Interface status, link light, and basic reachability-to-next-hop checks all reported healthy, since the cable and interface were genuinely up — only an OSPF neighbour-state check or an actual end-to-end traceroute would reveal that no adjacency, and therefore no route, existed. That is exactly why the fault was only found through manual traceroute troubleshooting at 12:30 IST rather than through any automated alert.

The organisational/process gap is visible directly in the change log: the Bengaluru (CHG-0238) and Pune (CHG-0241) OSPF changes both explicitly record 'verified adjacency' as part of the change description, but the Hyderabad change (CHG-0245) does not — it simply notes the Area ID error was introduced. This indicates the standard post-change

verification step was skipped or not enforced for that particular change, allowing a single transcription error to reach production and remain live for ten days without a second person or a validation step catching it.

D) SOLUTION DESIGN

- **Technical fix:** correct the Hyderabad router's Area ID to match the design intent shown in the topology diagram, and treat adjacency verification (confirming Full neighbour state on both ends, and confirming LSAs/routes are actually being learned) as a mandatory, recorded step for every OSPF change — matching what was already done correctly for Bengaluru and Pune.
- **Process-level safeguard:** require peer review of any Area ID (and other adjacency-critical parameter) change against the authoritative topology diagram before implementation, and make 'adjacency verified' a required, checked field in the change log template rather than optional free text — so a change cannot be marked Implemented without that evidence.

The move to multi-area OSPF is justified for NimbusEdge's backbone specifically because it directly addresses the weaknesses exposed by RIP and IGRP elsewhere in this incident: OSPF is link-state, so it converges quickly via SPF recalculation rather than periodic distance-vector updates and holddown timers; it has no artificial hop-count ceiling, giving it far more scalability headroom than RIP; and its hierarchical area design (Area 0 backbone plus regional areas) limits the scope of LSA flooding and routing-table size compared to a flat IGRP interconnect, which matters as NimbusEdge scales toward the planned Q3 2025 onboarding surge.

E) CLARITY

Before (fault state): Hyderabad router configured with an incorrect Area ID (CHG-0245), no recorded adjacency verification — no Full neighbour relationship with the Chennai Area 0 backbone router, despite the physical link (per the topology diagram) being up.

After (corrected state): Area ID corrected to match the topology diagram's design (CHG-0258, emergency change during the incident); adjacency confirmed Full, restoring Hyderabad-bound routes across the backbone.

Recommendation: Make adjacency verification a mandatory, recorded step for every OSPF change, backed by peer review against the topology diagram.

Q4. BGP — ISP Peering and Failover

PROBLEM STATEMENT

When the primary ISP's leased line failed, BGP's automatic failover to the secondary ISP did not trigger as designed, stalling all outbound client-facing traffic until a senior architect manually forced the path over — turning an ISP-side fault into a formal, client-facing SLA breach.

A) UNDERSTANDING OF THE CASE

NimbusEdge peers with two ISPs — ISP-A (primary) and ISP-B (secondary) — over BGP for redundant internet access, per the Network Topology Diagram. The RCA-related Change Log shows CHG-0249 (2024-03-11), just a week before the incident, reviewing LOCAL_PREF values for ISP-A/ISP-B outbound preference but explicitly noting it was 'not re-tested end-to-end' — the policy was adjusted but never exercised through an actual failover.

On the incident day, the Incident Report records that at 14:50 IST, ISP-A reported a fiber cut affecting NimbusEdge's leased line, and automatic failover to ISP-B did not occur — all outbound client-facing traffic stalled. This forced manual BGP intervention: at 15:40 IST a senior network architect manually forced traffic onto ISP-B, with the ISP-B path confirmed active by 16:20 IST. The Client SLA Breach Notice confirms the downstream consequence — this outage window (part of the broader 09:42–18:55 IST incident) contributed to a breach of the contracted 99.9% monthly uptime SLA, triggering formal client notification and service credits.

B) APPLICATION OF CONCEPTS

BGP selects a best path using ordered attributes including **LOCAL_PREF** (a locally significant value expressing an AS's own outbound preference — higher is preferred, and it overrides AS-PATH length in the decision order), AS-PATH (shorter is generally preferred; can be artificially lengthened via prepending to de-prioritise a path), and MED (a hint to a neighbouring AS about which of multiple entry points to prefer — lower is preferred, and it typically applies to inbound path selection between two connections to the same neighbour AS).

The likely policy misconfiguration here is that **LOCAL_PREF** continued to favour the ISP-A path as NimbusEdge's outbound preference even once ISP-A's route became unusable, because the **LOCAL_PREF**/AS-PATH policy — as CHG-0249 itself notes — had been reviewed on paper but never validated in a live failover. In a correctly designed and tested policy, the loss of the ISP-A route (or a health signal reflecting the fiber cut) should trigger BGP's best-path algorithm to recalculate and select the ISP-B path as the new best route automatically, since it would then be the only, or the more preferred, remaining option. That automatic recalculation did not happen as expected, consistent with an outbound preference/detection policy that had not been proven against a real failure of this kind.

C) ANALYSIS

An untested failover policy is higher risk in BGP specifically than in an interior gateway protocol because BGP governs the single external point of contact for an entire organisation's client-facing traffic — an IGP fault (like the RIP or OSPF issues elsewhere in this incident) is typically contained to one internal segment or site, whereas a BGP outbound policy failure at the ISP edge affects every client and every site simultaneously, as it did here (all outbound client-facing traffic stalled, not just one office). BGP failover paths are also exercised far less often than internal routing changes — a primary ISP rarely fails — so a policy error can sit completely dormant and 'correct-looking' for months or years until the one moment it is actually needed, which is exactly the pattern CHG-0249 shows: reviewed and adjusted on paper, never tested end-to-end.

The business consequence chain is direct and traceable through the provided documents: an untested **LOCAL_PREF**/AS-PATH policy → automatic BGP failover did not trigger at 14:50 IST → all outbound client-facing traffic stalled until manual intervention restored service via ISP-B by 16:20 IST → roughly 90 additional minutes added to an already-severe outage → contributing to a total incident duration of about 9 hours 13 minutes across six enterprise clients → breach of the contracted 99.9% monthly SLA → the formal SLA Breach Notice issued to affected clients, with associated service credits and reputational exposure.

D) SOLUTION DESIGN

- **Corrected policy:** configure **LOCAL_PREF** and AS-PATH so that ISP-B is automatically preferred the moment the ISP-A route is withdrawn or a health signal indicates ISP-A is unusable — for example, pairing the outbound policy with fast failure detection (BFD) or a tracked health check tied to route advertisement, rather than relying solely on the eventual withdrawal of the BGP route itself, which can lag a physical fiber cut.
- **Testing/validation cadence:** scheduled, planned failover drills (e.g., quarterly, in a maintenance window) that actually force traffic across to ISP-B and confirm automatic reconvergence — closing the exact gap CHG-0249 left open — plus a change-process rule that any BGP outbound-policy change must include an end-to-end failover test as a condition of being marked Implemented, mirroring the OSPF adjacency-verification recommendation in Q3.

BGP is the necessary and correct choice for NimbusEdge's dual-ISP redundancy — rather than an interior gateway protocol — because ISP-A and ISP-B are separate autonomous systems, and only an exterior gateway protocol like BGP provides policy-based control (via **LOCAL_PREF**, AS-PATH, and MED) over which of multiple external paths is preferred for inbound and outbound traffic across AS boundaries. An IGP has no mechanism for expressing preference between two different providers' autonomous systems; BGP is what makes controlled, redundant multi-homing to two ISPs possible at all. The incident is not evidence that BGP was the wrong tool — it is evidence that the policy running on top of BGP was never proven under real failure conditions.

E) CLARITY

Before (fault state): *LOCAL_PREF*/AS-PATH policy continues to favour ISP-A even after its route becomes unusable; no automatic shift to ISP-B when the fiber cut occurs at 14:50 IST.

After (corrected state): *LOCAL_PREF* policy re-tuned so ISP-B is automatically preferred the moment ISP-A becomes unreachable, ideally backed by fast failure detection; validated through a live failover test rather than a paper review.

Recommendation: Redesign the outbound preference policy for automatic reconvergence, and institute a recurring, mandatory end-to-end failover test before any related change is closed.

Note: All facts above are drawn directly from the four supplied source documents (Incident Report NOC-INC-2024-031, Network Topology Diagram, NOC Change Log Feb–Mar 2024, and Client SLA Breach Notice). Where the RCA_Report_Q1_2024.docx is referenced by these sources but was not itself supplied, the corresponding points have been drawn only from what the Incident Report and Change Log independently confirm.